LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	12
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Ratio meter system, thermoelectric thermometers
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Ratio meter system, thermoelectric thermometers
5. Specific Objectives	15.1.5.2 Trainee will be able to explain ratio meter type temperature indicator.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of schematic diagrams; Instructor supervision of diagram interpretation; Safety: No live electrical testing on board.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a temperature indicator?
(b) How does a resistance change with temperature variation?
(c) Can you name one type of temperature sensor used in aircraft?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Principle of Ratio Meter operation • Bridge circuit configuration • Deflection mechanism explanation • Calibration and zero adjustment • Advantages over other systems	L & Ds	PPT, IB	15 min
(b)	• Thermoelectric thermometer principle • Seebeck effect explanation • Thermocouple junction types • Signal generation and measurement • Applications in aircraft systems	L & Ds	PPT, PDF	20 min
(c)	• Comparison of Ratio Meter and Thermoelectric systems • Maintenance requirements • Troubleshooting common faults • Operational limitations • Integration with aircraft warning systems	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the ratio meter indicates temperature changes.
(b) Describe the Seebeck effect in the context of thermoelectric thermometers.
(c) List two advantages of the ratio meter system over direct resistance measurement.

12. Home Assignment and Follow Up	Prepare a brief note comparing the operating principles of the ratio meter system and thermocouple systems, highlighting their respective applications in aircraft power generation and distribution.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Thermocouple principle, thermocouple materials and combinations. Types of thermocouples, Thermocouple harness assemblies. Cold junction temperature compensation	Refer to the master syllabus reference list	LP 13

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Captain, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min